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MILITARY SPECIFICATION

LAPPERS, POPPET VALVE, HAND POWERED

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

★ 1.1 Scope. This specification covers hand powered lappers of the oscillating, progressively or irregularly advancing type, used primarily for lapping of automatic engine valves and seats.

★ 1.2 Classification. Lappers shall be of the following grades, as specified (see 6.2):

Grade A—Heavy Duty
Grade B—Light Duty

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATION

MILITARY

MIL-H-15424—Hand Tools; Packaging of.

STANDARDS

MILITARY

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.
MIL-STD-130—Identification Marking of U.S. Military Property.

(Copies of the specifications, standards, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS PUBLICATIONS

Handbook H28—Screw Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D.C.).

3. REQUIREMENTS

★# 3.1 Preproduction sample. When specified (see 6.2), before production is commenced, a sample of the tool covered by this specification shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.2. The approval of the preproduction sample authorizes the commencement of production, but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production items.

FSC 5120

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★# 3.2 Illustrations. Figure 1 is for the purpose of illustration only, and is not intended to exclude lappers otherwise conforming to this specification.

3.3 Standard product. The lapper shall be a current standard product of the manufacturer, except for any modification specified herein.

★# 3.4 Material. The materials specified shall conform to the requirements specified hereinafter. When materials are not definitely specified, they shall be of the quality normally used for the purpose in commercial practice.

★# 3.5 Design and construction. The lapper shall consist essentially of a housing, crank arm, valve lifting spring, extension drive stem, drive points, gear train or mechanism, ball thrust bearing and rubber suction cups. The lappers shall be similar to figure 1. The lapper shall produce an oscillation of the valve around its axis with progressive or irregular advancement. Lappers shall be so constructed that no assembled part will become detached during normal use. All moving parts shall operate without binding or looseness that would impede proper function of tool.

3.5.1 Grade A, heavy duty lapper shall be capable of withstanding 500,000 revolutions when tested as specified in 4.4.3.

3.5.2 Grade B, light duty lapper shall be capable of withstanding 90,000 revolutions when tested as specified in 4.4.3.

3.5.3 *Housing.* The housing shall be of high quality cast iron or drawn steel. The housing shall be of two half sections or one section with a cover. In either case, the half sections or cover plate shall be assembled securely to minimize both the loss of light density grease and the entrance of foreign particles. The assembly shall allow ready access for lubrication and removal and replacing of parts. The housing shall properly support the inclosed mechanism. A handle or knob which affords a comfortable grip shall be attached to or integral with the top side of the housing.

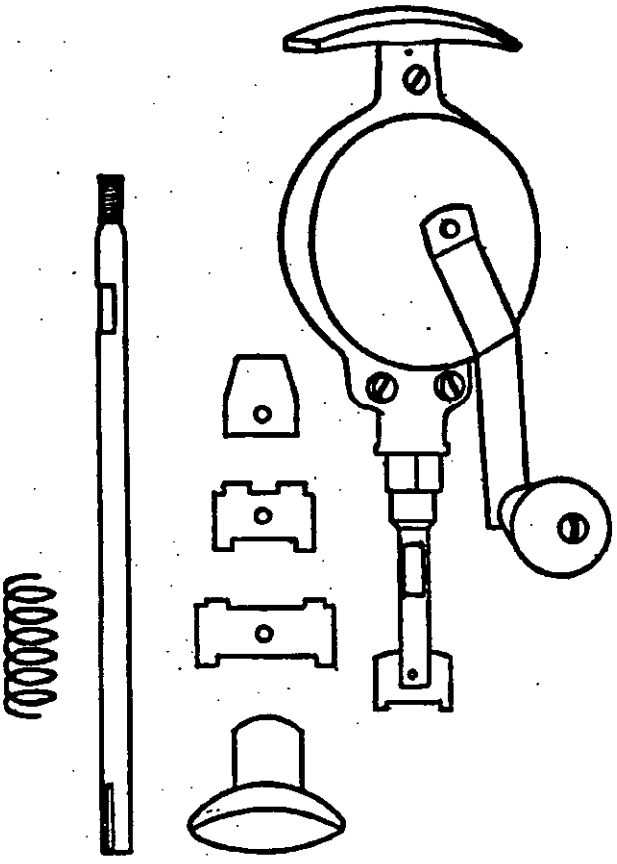


FIGURE 1. *Oscillating valve lapper, grades A and B.*

★ 3.5.4 *Crank Arm.* The crank arm shall be of a length and shape to assure clearance for hands in operating, and have sufficient strength to withstand intended usage. One end of the crank arm shall provide means for attaching to crank shaft. The other end of the arm shall have a handle (knob) affixed to rotate on a bolt, rivet or pin of the head and shoulder type.

★ 3.5.5 *Valve lifting spring.* The lifting spring shall be helical, or hard drawn spring wire, oil tempered wire or piano wire. The spring shall be 2¼ inches plus or minus ⅛ inch long by 1 inch plus or minus 1/16 inch in diameter. The helical spring may be straight-wound or the ends may be conical in shape.

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★ 3.5.6 *Extension drive stems.* Each lapper shall be furnished with a long drive stem a minimum of 7 inches long and maximum of 8 inches long and a short drive stem a minimum of 3 inches long, a maximum of 4 inches long. One end of the drive stem shall be threaded for attachment to the driving spindle. The opposite end of the drive stem shall be slotted and drilled for attachment of drive points.

★ 3.5.7 *Drive points.* Drive points, in the amount and sizes normally furnished by the manufacturer, shall be included with the lapper. The points may be either one piece universal blade type or individual units and shall contain a hole to allow for attachment to the stem with a cotter pin. When specified (see 6.2), special drive points shall be supplied.

★ 3.5.8 *Gear train or mechanism.* Gears and pinions, if applicable, shall be secured to crank and drive shafts by a positive means to assure proper alignment, rotation and meshing. In the grade A lapper, the relation between the spacing and number of teeth of the driving and driven gears or the capability of the operating mechanism shall be such that the sum total of the forward and reverse rotation of the drive stem will total between 450 and 675 degrees for each revolution of the crank. The grade B lapper shall conform to the above requirements except that the sum of both forward and reverse rotation of the drive stem shall total between 360 and 720 degrees for each revolution of the crank.

★ 3.5.9 *Ball thrust bearing.* A ball thrust bearing shall be used to minimize friction and wear while the vertical drive shaft is rotated under pressure. Construction of bearing shall be such that the balls will not be disturbed when removing rubber suction cup from the drive stem.

★ 3.5.10 *Rubber suction cups.* Each lapper shall be equipped with ten, vacuum type, rubber suction cups. Five suction cups shall be 1¼ inches in diameter (approximately), and five suction cups shall be 1¾ inches in diameter. They shall be of conventional shape, designed to force fit the extension drive stem or be secured by cotter pins to securely hold valves not equipped with spanner holes or slot in the valve head. The rubber cups shall be of a neoprene compound having a hardness range of 55–A to 65–A Durometer (Shore).

★ 3.5.11 *Lubrication.* The bearings and gears shall be lubricated with a light density grease to render the lapper ready for immediate use.

★ 3.5.12 *Weight.* The weight of a lapper with a cast iron housing shall be 3½ to 5 pounds for grade A, and 2 to 3 pounds for grade B. The weight of a lapper with a steel housing shall be 1½ to 2½ pounds for grade A and B.

★ 3.6 Surface finish. Castings shall be thoroughly cleaned and free of sand, fins, and sprues. All parts shall be thoroughly cleaned and free of scale, dirt or other harmful or extraneous materials. All external surfaces and edges shall be free of burrs, fins and sharp edges. The lappers shall show no evidence of rust.

★ 3.7 Painted or coated housing surfaces. All surfaces of the housing normally painted as in manufacturer's commercial practice shall be furnished with at least one coat of enamel or lacquer; otherwise they shall be protected by a dense phosphate or oxide coating followed by a coating of oil, grease or compound preservative.

★ 3.8 Corrosion-resisting surfaces. All parts and surfaces, except the housing, which are subject to corrosion shall be protected by cadmium plate, zinc plate, or by a dense phosphate or oxide coating, followed by a coating of oil, grease or compound preservative.

★ 3.9 Threads. All threaded parts shall be in accordance with Handbook H–28.

★ 3.10 Workmanship. Lappers shall conform to the quality and grade of product established by this specification. The lapper shall be clean, well made and properly assem-

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bled. Parts and assemblies shall not be damaged or impaired in any way.

3.11 Marking. Marking shall be in accordance with MIL–STD–130.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction sample inspection. When a preproduction sample is required it shall be examined for defects in 4.3.3.1, dimensions and weight, and tested as specified in 4.4.2, 4.4.3, and 4.4.4.

★ 4.3 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be made in accordance with the provisions set forth in MIL–STD–105, except where otherwise indicated hereinafter.

★ 4.3.1 *Inspection of components and materials.* In accordance with 4.1 above, the supplier is responsible for insuring that components and materials used were manufactured, tested and inspected in accordance with this specification.

★★ 4.3.2 *Intermediate testing.* Rubber suction cups shall be tested for hardness to determine compliance with 3.5.10 and test in 4.4.1. A lot shall consist of rubber suction cups treated as a batch or under the same process or control. The sample unit shall be one rubber suction cup. The inspection level shall be S–2 with an acceptable quality level (AQL) of 2.5 defects, expressed in terms of defects per hundred units.

4.3.3 *Inspection of the end item.* The inspection lot shall be the number of valve lappers of the same grade offered for acceptance at one time. The sample unit shall be one completely assembled valve lapper.

4.3.3.1 *Visual examination.* The examination of the end item shall be in accordance with the classification of defects and acceptable quality levels (AQL's) set forth below. The inspection level shall be level II with an AQL of 2.5 defects, expressed in terms of defects per hundred units.

Examine	*Defect*
Finish	Not free of sand, dirt, scale or extraneous materials.
	Any burrs, fins or sharp edges.
	Not corrosion resistant as specified.
	Evidence of rust.
Paint or coating	Chipped, peeling, blistered, scratched or foreign matter imbedded.
	Area not coated. Coating not as specified.
Design, construction and workmanship.	Part missing or not specified type.
	Fractured, split, cracked or otherwise malformed or distorted.
	Functional component that requires abnormal force to operate.
	Component loose or not in proper alignment.
Marking identification	Missing, illegible, or incomplete.

4.3.3.2 *Dimensional and weight examination.* Examination shall be made of the end item to determine compliance with dimensional and weight requirements. Any dimension or weight not within the specified requirements shall be classified a defect. The inspection level shall be S–3 with an AQL of 2.5, defects, expressed in terms of defects per hundred units.

4.3.3.3 *End item test.* Sample lappers shall be tested as specified in 4.4.2 and 4.4.4 to determine compliance with 3.5 and 3.5.2. The inspection level shall be S–2 with an AQL of 1.5 defects, expressed in terms of defects per hundred units. When no preproduction sample is required, one production unit from the initial production lot shall be tested as specified in 4.4.3. Any noncompliance shall be cause for rejection of the lot.

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4.3.4 *Examination for preparation for delivery.* An examination shall be made to determine compliance with preservation, packaging, packing and marking requirements in accordance with section 5 and MIL-H-15424.

4.4 Test.

4.4.1 *Test for hardness of rubber.* Test shall be made in accordance with Method 3021 of Federal Test Method Standard No. 601.

4.4.2 *Load test.* Each sample lapper shall be operated by turning the crank 20 revolutions. In order to simulate the action of lapping a valve and valve seat of an automobile, the drive points or suction cup shall engage the top side of a valve resting in a valve seat. The crank shall be turned 10 revolutions without any endwise or vertical load upon the unit. The lapper crank shall then be turned 10 revolutions with an endwise or vertical load of 10 pounds added to the unit. During the test and after completion, there shall be no evidence of binding or failure to function properly.

4.4.3 *Endurance test.* With the crank removed and a power source connected to the crank shaft, the lapper shall operate at 200 revolutions per minute plus or minus 20 percent, with a torque load between 2 and 3 inch pounds applied to the drive stem. There shall be no permanent deformation to the gear and pinion teeth or the driving mechanism as the results of this test and the requirements specified in 3.5.1 and 3.5.2.

4.4.4 *Gear train or driving mechanism test.* The crank handle shall be in the top most position and a mark placed on the rubber suction cup attached to drive stem. The lapper shall be placed perpendicular to a 360 degree protractor. The mark on the rubber cup shall be placed at zero degrees of the protractor. One full revolution of the crank handle shall produce the required angular rotation specified in 3.5.8.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing for levels A, B, or C shall be as specified in MIL-H-15424 (see 6.2).

5.2 Marking. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Grades A and B lappers are intended for use in maintenance which requires the lapping of valves and valve seats of automobile engines. They are satisfactory for valves and valve seats requiring a normal amount of lapping, but not especially recommended for valve and valve seats which are badly pitted, burned, or scored, unless the valves and valve seats cannot be replaced with new parts or that power tools are not available.

6.1.1 *Grade A lappers.* Grade A lappers are considered to be appropriate for use in maintenance shops which may encounter heavy workloads of valve and valve seat lapping during engine repairs. They are considered to have satisfactory service life of longer duration than Grade B.

6.1.2 *Grade B.* Grade B lappers are considered to be appropriate for intermittent use and are suitable for inclusion in tool boxes for maintenance of automobile engines.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Grade required (see 1.2).
(c) Whether preproduction sample is required (see 3.1).
(d) When special drive points are required (see 3.5.7).
(e) Selection of the applicable levels of packaging and packing (see 5.1).

6.3 The outside margins of this document have been marked to indicate where changes from the previous issue have been

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